AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

ANNUAL EXAMINATIONS (THEORY) 2024

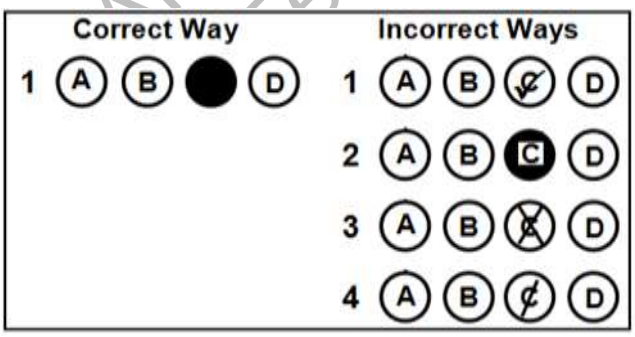
General Mathematics Paper I

Time: 1 hour 20 minutes Marks: 45

AKU-EB Annual Examinations 2024 Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 45 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a simple calculator if you wish.

List of Formulae

NOTE:

- The symbols have their usual meanings.

Financial Mathematics

$$I = PT \times \frac{R}{100}$$

Basic Statistics

$$\overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{n} \quad \text{or} \quad \overline{X} = \frac{\sum fx}{\sum f} \qquad \text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h$$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \qquad \sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Quadratic Equation

$$ax^2 + bx + c = 0,\ a \neq 0 \qquad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \text{Disc} = b^2 - 4ac$$

Sequence and Series

$$a_n = a + (n-1)d \qquad a_n = ar^{n-1} \qquad AM = \frac{a+b}{2} \qquad GM = \pm\sqrt{ab}$$

Area and Volumes

$$\text{Area of a circle} = \pi r^2 \qquad \text{Area of a triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Volume of a cube} = l^3 \qquad \text{Volume of a cuboid} = l \times b \times h \qquad \text{Volume of a sphere} = \frac{4}{3} \times \pi r^3$$

$$\text{Volume of a cone} = \frac{1}{3} \times \pi r^2 \times h \qquad \text{Volume of a cylinder} = \pi r^2 \times h$$

Introduction to Coordinate Geometry

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad M.P = \left(\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2}\right)$$

Algebraic Formulae

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^2 - b^2 = (a+b)(a-b) \qquad (a+b)^2 = a^2 + 2ab + b^2 \qquad a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \qquad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

AKU-EB Annual Examinations 2024 for Teaching & Learning only

Use the given information to answer Q.1 and Q.2.

Sultan plans to buy a new car on instalments from a bank. The payment details of the bank are given in the table.

S. No.	Payment	Detail
1	Price of the Car	Rs 2,500,000
2	Interest of the Bank	20% of the Price of the Car
3	Down Payment	25% of (Price of the Car + Interest of the Bank)
4	Bank Processing Fee	Rs 2,500 (At the beginning of the Purchase)
5	Instalment Period	3 Years
6	Number of Equal Monthly Instalments	36

1. The down payment for the car he has to make is Rs

   A. 752,400
   B. 750,000
   C. 627,500
   D. 625,500

2. The monthly instalment of the car will be Rs

   A. 48,611
   B. 62,500
   C. 83,333
   D. 104,236

3. If Rs x becomes Rs $3x$ at a simple interest rate of 6 % per annum, then the duration of the investment will be

   A. $\frac{2}{100}$.
   B. $\frac{3}{100}$.
   C. $\frac{100}{3}$.
   D. $\frac{100}{2}$.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

4. Ahmer and his friends are planning a trip to Europe from North America. Therefore, they need to convert US dollars (USD) into euros (EUR). How many euros in rounded figure will they receive for 150 US dollars?

   (**Note:** The current exchange rate is 1 EUR = 1.09 USD.)

   A. 79 EUR
   B. 138 EUR
   C. 164 EUR
   D. 285 EUR

5. Maria is working in a factory outlet. Her daily wage is Rs 1,000 and she gets 5% commission on the sales she makes. If she makes sales of Rs 250,000 in five days, then her total earnings for the five days will be

   A. Rs 12,500
   B. Rs 12,750
   C. Rs 17,500
   D. Rs 17,750

6. Akram's monthly taxable income is Rs 350,000. If he pays a monthly tax of Rs 50,000, then the monthly tax rate is

   A. 10.28%.
   B. 12.67%.
   C. 14.28%.
   D. 16.67%.

7. The given table shows the rate per unit of electricity charges for different slabs.

0 - 100 units	101 - 200 units	201 - 500 units	501 and more units
Rs 6.2/ unit	Rs 12.6/ unit	Rs 17.1/ unit	Rs 22.4/ unit

   The electricity consumption of Jamila's house is 350 units in a certain month. The accumulated electricity charges she will pay for this month is

   (**Note:** The exemption will be provided based on the specified slabs according to the total units consumed)

   A. Rs 3,995
   B. Rs 4,445
   C. Rs 5,085
   D. Rs 5,985

AKU EB Annual Examinations 2024 for Teaching & Learning only

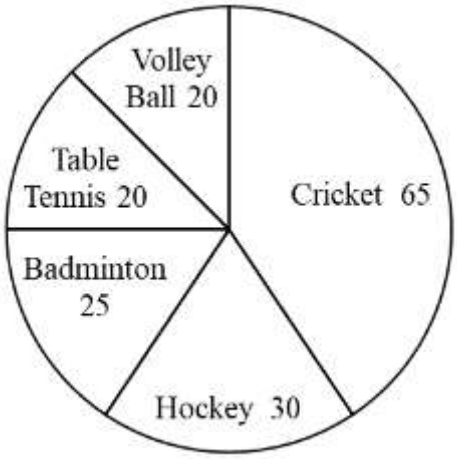
8. Given is the frequency distribution table of weights, in grams, of 50 apples.

Weight of Apples (in grams)	Number of Apples
30 − 34	4
35 − 39	8
40 − 44	16
45 − 49	12
50 − 54	10

The lower class limit of the class interval which has frequency 12 is

A. 44.5
B. 45.0
C. 45.5
D. 49.0

9. A survey is conducted to obtain data about the favourite sport of 160 students of class X. The result of the survey is represented by the given pie-chart.



The angle of the sector representing hockey is

A. 67.5^{o}
B. 30.0^{o}
C. 18.8^{o}
D. 13.3^{o}

10. If the mean of $2a-1,\ 2a+2,\ 2a+3$ and $2a+5$ is 31, where a is a positive number, then the value of a will be

A. 2.75
B. 5.0
C. 11.0
D. 14.38

for -EB ...nations 2024 ...y & Learning only

11. For the given data 3, 3, 5, 5, 5, 7, 7 the correct relationship between mean, median and mode is

A. mean = median = mode.
B. mean > median = mode.
C. mode > mean > median.
D. median = mean > mode.

12. The given table shows the age group of staff working in a certain company.

Age (in Years)	26 - 30	31 - 35	36 - 40	41 - 45	46 - 50
Number of Employees	6	15	22	7	4

The number of staff below or equal to 40 years of age is

A. 15
B. 21
C. 22
D. 43

13. If $\frac{\sum x^2}{n} = \frac{110}{3}$ and $\overline{X}^2 = 36,$ then the variance will be

A. $-\frac{74}{3}.$
B. $\frac{74}{3}.$
C. $-\frac{2}{3}.$
D. $\frac{2}{3}.$

14. The highest common factor (HCF) of the polynomials $x^2 + 6x + 9$ and $x^3 + 9x^2 + 27x + 27$ is

A. 1
B. x
C. $x + 3$
D. $x^2 + 6x + 9$

15. The product of the least common multiple (LCM) and highest common factor (HCF) of two polynomials is $(x^3 + 8)(x + 2)^2.$

If one of the polynomials is $(x + 2)^2$, then the other polynomial will be

A. $x^3 + 8$
B. $(x + 2)^2$
C. $(x^2 + 2x + 4)$
D. $(x^2 + 4x + 4)$

AKU-EB Annual Examinations 2024 for Teaching & Learning only

16. The least common multiple (LCM) of $(x+a)(x+b)$ and $(2x+2a)(x+c)$ is

A. $(x+a)$.
B. $2(x+a)$.
C. $(x+a)(x+b)(x+c)$.
D. $2(x+a)(x+b)(x+c)$.

17. The positive square root of the expression $\frac{(x^2-1)(x+1)}{(x-1)^2(x-1)}$ is

A. $x+1$
B. $x-1$
C. $\frac{x-1}{x+1}$
D. $\frac{x+1}{x-1}$

18. On simplification of $\frac{1}{x-2}+\frac{1}{x+2}$, we get

A. $\frac{2x}{x^2-2}$.
B. $\frac{4}{x^2-2}$.
C. $\frac{2x}{x^2-4}$.
D. $\frac{4}{x^2-4}$.

19. The solution set of the equation $2(x-3)+4=4$ is

A. $\{1\}$.
B. $\left\{\frac{3}{2}\right\}$.
C. $\{3\}$.
D. $\{5\}$.

20. The value of x in the given equation $\sqrt{\frac{x-a}{a}}=0$ is

A. $\{0\}$.
B. $\{a\}$.
C. $\{\pm a\}$.
D. $\{2a\}$.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

21. Aleem is three years older than Shariq. Three years ago, Aleem was twice as old as Shariq. Then the present age of Shariq is

A. 3 years.
B. 6 years.
C. 9 years.
D. 12 years.

22. For the equation $|x+1|=1$, the values of x will be

A. -2 and 0
B. -2 and 2
C. 1 and 0
D. 0 and 2

23. For x, a, b and c are all real numbers, 'if $x+a>x+b$ and $x+b>x+c$, then $x+a>x+c$'.

This property is called

A. additive property.
B. transitive property.
C. inequality property.
D. trichotomy property.

24. The solution set of $-\frac{1}{x}<-1$, where $x \in R$, will be

A. $\{x \mid x \in R, x>1\}$.
B. $\{x \mid x \in R, x<1\}$.
C. $\{x \mid x \in R, x<-1\}$.
D. $\{x \mid x \in R, x>-1\}$.

25. The solution set of the quadratic equation $2x^2-8=0$ is

A. $\{2\}$.
B. $\{4\}$.
C. $\{-2, 2\}$.
D. $\{-4, -4\}$.

26. The standard form of the quadratic equation $(x-a)(x-b)=0$ is

A. $x^2+(a-b)x+ab=0$
B. $x^2-(a-b)x-ab=0$
C. $x^2-(a+b)x-ab=0$
D. $x^2-(a+b)x+ab=0$

AKU-EB Annual Examinations 2024 for Teaching & Learning only

27. The solution set of the quadratic equation $2x^2 + x + 2 = 0$ is

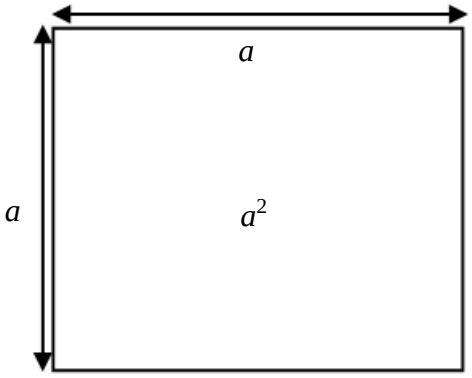
A. $\left\{\frac{1 \pm \sqrt{17}}{4}\right\}$.

B. $\left\{\frac{-1 \pm \sqrt{17}}{2}\right\}$.

C. $\left\{\frac{-1 \pm \sqrt{-15}}{4}\right\}$.

D. $\left\{\frac{1 \pm \sqrt{-15}}{2}\right\}$.

28. Consider a square paper with length a cm. If the length is reduced by 2 cm on one side and increased by 1 cm on the other side of the square, the area of the rectangle formed is 18 cm^2, then the value of a will be



A. 2
B. 3
C. 5
D. 6

29. The 7th term of the sequence – 3, – 4, – 5, – 6, … is

A. –7
B. –8
C. –9
D. –10

30. If the common difference of the arithmetic sequence $-3, -p, -q, -9, \ldots\ldots$ is –2, then the value of p is

A. –5
B. –1
C. 5
D. 1

Annual Examinations 2024 for Teaching & Learning only

31. If an arithmetic sequence is defined as 1, b, $\frac{1}{2}$,…, then b is equal to

A. $-\frac{1}{2}$.

B. $\frac{1}{2}$.

C. $\frac{3}{4}$.

D. $\frac{3}{2}$.

32. The geometric mean between $(a-b)^2$ and $\frac{1}{a-b}$ is

A. $\pm\sqrt{a-b}$.

B. $\pm(a-b)$.

C. $\pm\frac{a-b}{2}$.

D. $\pm\sqrt{(a-b)^3}$.

33. The first term of a geometric sequence is $\frac{5}{2}$ and common ratio is $\frac{3}{2}$. The 4th term of the sequence is

A. $\frac{135}{16}$.

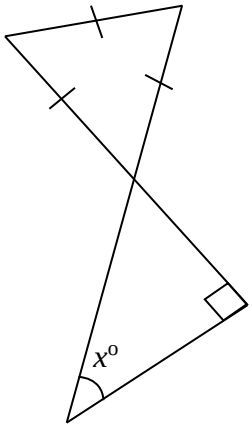
B. $\frac{405}{32}$.

C. $\frac{475}{16}$.

D. $\frac{2375}{32}$.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

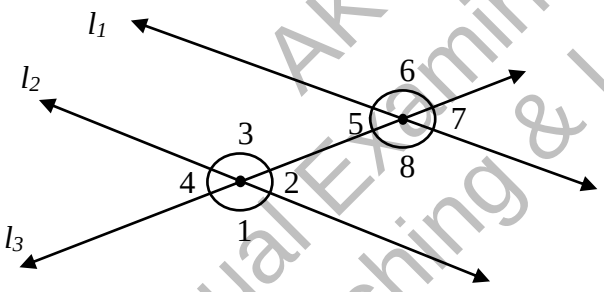
34. In the given diagram, the value of x is equal to



NOT TO SCALE

A. 30°
B. 120°
C. 150°
D. 270°

35. In the given diagram, a transversal l_3 intersects the two parallel lines l_1 and l_2.



NOT TO SCALE

If $\angle 5 = x°$, then $\angle 1$ will be equal to

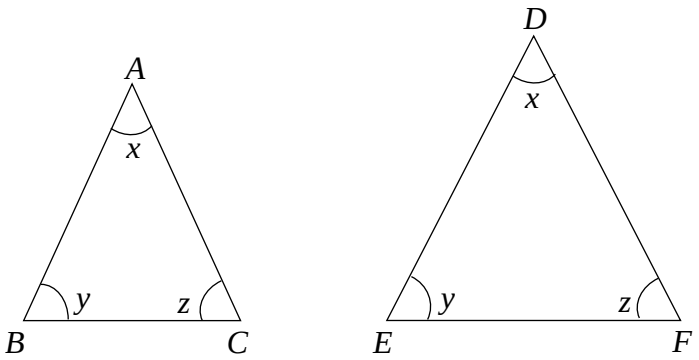
A. $x°$.
B. $(90 - x)°$.
C. $(90 + x)°$.
D. $(180 - x)°$.

36. If one of the sides of a parallelogram is $\frac{s}{2}$ units, then its opposite side will be

A. $\frac{2}{s}$ units.
B. $\frac{s}{2}$ units.
C. $2s$ units.
D. s units.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

37. Consider the given correspondence of $\Delta ABC \cong \Delta DEF$.



NOT TO SCALE

The CORRECT relation between the sides of two triangles is

A. $\frac{AB}{BC} = \frac{DE}{EF}$.

B. $\frac{AB}{DE} = \frac{AC}{EF}$.

C. $\frac{AB}{BC} = \frac{DE}{DF}$.

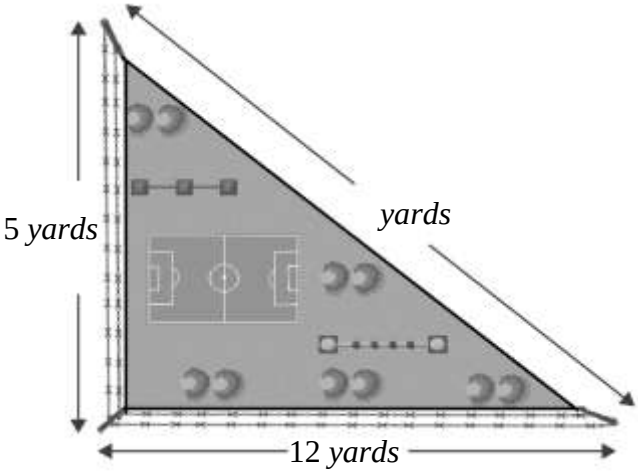
D. $\frac{AB}{DE} = \frac{AB}{BC}$.

38. If the base and the altitude of a parallelogram are 8 cm and $x - 2$ cm respectively, then its area will be

A. $8(x - 2)$ cm^2
B. $4(x - 2)$ cm^2
C. $2(x + 6)$ cm^2
D. $\frac{(x+6)}{2}$ cm^2

AKU-EB Annual Examinations 2024 for Teaching & Learning only

39. A sports centre is in the shape of a right angled triangle. Two of its sides are covered with net as shown in the given figure. The third one is required to be covered.

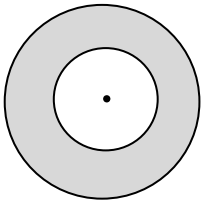


NOT TO SCALE

The length of the net required to cover the remaining side will be

A. $\sqrt{119}$ cm.
B. $\sqrt{169}$ cm.
C. 169 cm.
D. 289 cm.

40. The shaded area enclosed by concentric circles of radii 3 cm and 4 cm, as shown in the given figure, is



A. π cm^2
B. π^2 cm^2
C. 7π cm^2
D. $7\pi^2$ cm^2

41. If the diametre of a sphere is d m, then its volume will be

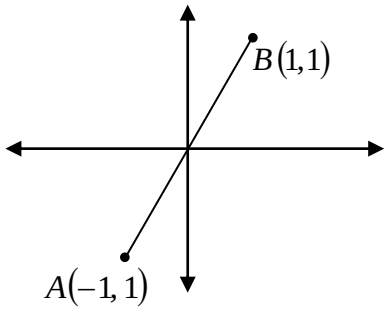
A. $\frac{2}{3}\pi d^3$ m^3
B. $\frac{4}{3}\pi d^3$ m^3
C. $\frac{1}{6}\pi d^3$ m^3
D. $\frac{1}{2}\pi d^3$ m^3

42. A steel cylinder has a radius of 10 cm and its volume is 1000π cm^3. The height of the cylinder is

A. 0.01 cm.
B. 0.1 cm.
C. 10 cm.
D. 100 cm.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

43. In the given diagram, the distance between point *A* and point *B* is



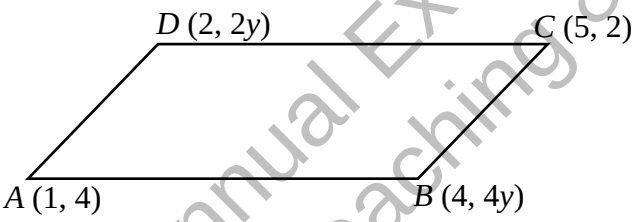
NOT TO SCALE

A. 2
B. $2\sqrt{2}$
C. 4
D. 8

44. The points *A* (0, 5), *B* (0, 3) and *C* (4, 0) are the vertices of a/ an

A. right angled triangle.
B. equilateral triangle.
C. isosceles triangle.
D. scalene triangle.

45. The vertices of a parallelogram are shown in a given figure.



NOT TO SCALE

If the point of intersection of *BD* and *AC* is (3, 3), then the value of *y* is

A. 0.5
B. 1
C. 3
D. 1.8

AKU-EB Annual Examination- 2024 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only